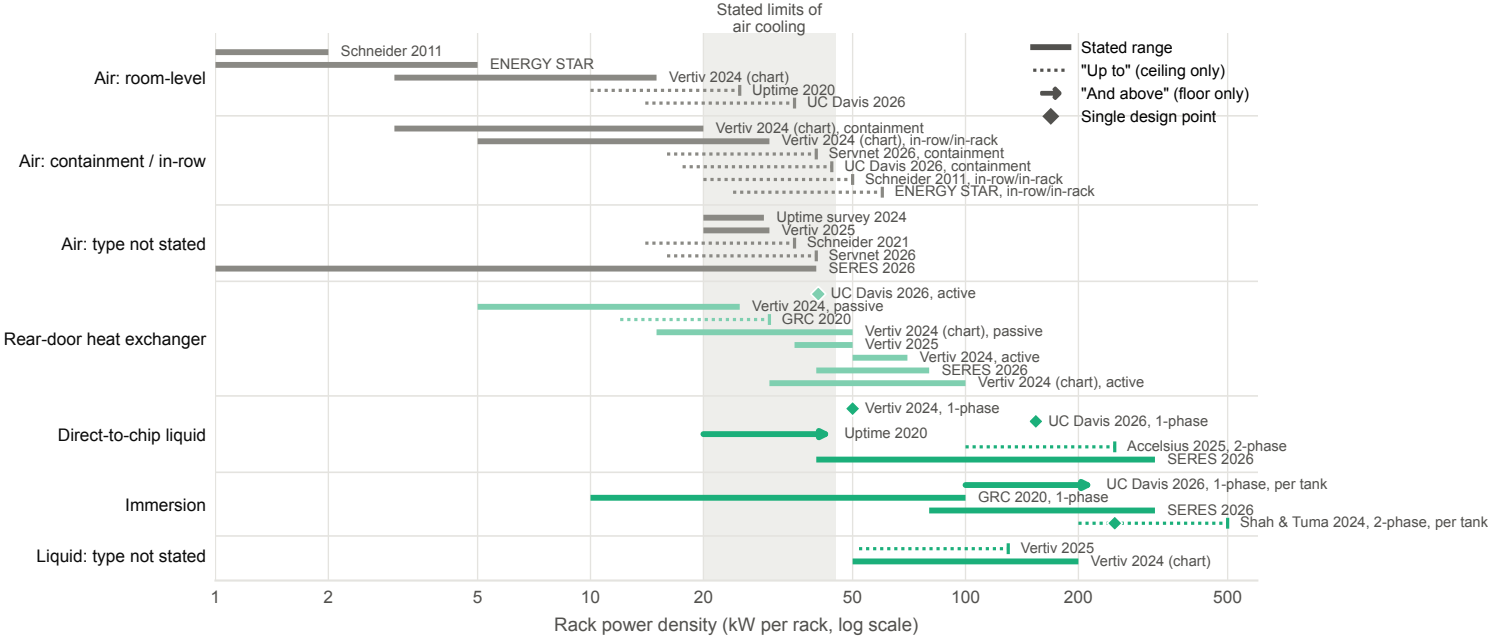
Stated limits of air cooling
Stated range
"Up to" (ceiling only)
"And above" (floor only)
Single design point
Air: room-level
Schneider 2011
ENERGY STAR
Vertiv 2024 (chart)
Uptime 2020
UC Davis 2026
Air: containment / in-row
Vertiv 2024 (chart), containment
Vertiv 2024 (chart), in-row/in-rack
Servnet 2026, containment
UC Davis 2026, containment
Schneider 2011, in-row/in-rack
ENERGY STAR, in-row/in-rack
Air: type not stated
Uptime survey 2024
Vertiv 2025
Schneider 2021
Servnet 2026
SERES 2026
Rear-door heat exchanger
UC Davis 2026, active
Vertiv 2024, passive
GRC 2020
Vertiv 2024 (chart), passive
Vertiv 2025
Vertiv 2024, active
SERES 2026
Vertiv 2024 (chart), active
Direct-to-chip liquid
Vertiv 2024, 1-phase
UC Davis 2026, 1-phase
Uptime 2020
Accelsius 2025, 2-phase
SERES 2026
Immersion
UC Davis 2026, 1-phase, per tank
GRC 2020, 1-phase
SERES 2026
Shah & Tuma 2024, 2-phase, per tank
Liquid: type not stated
Vertiv 2025
Vertiv 2024 (chart)
1
2
5
10
20
50
100
200
500
Rack power density (kW per rack, log scale)